

GrowthPilot AI

Customer operations, explainable inactivity risk, and governed marketing action

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Marketing decisions fail when the evidence trail is fragmented

01 Customer context

CRM notes and consent are detached from orders and value.

02 Commercial truth

Inventory, returns and revenue definitions drift across files.

03 Paid media

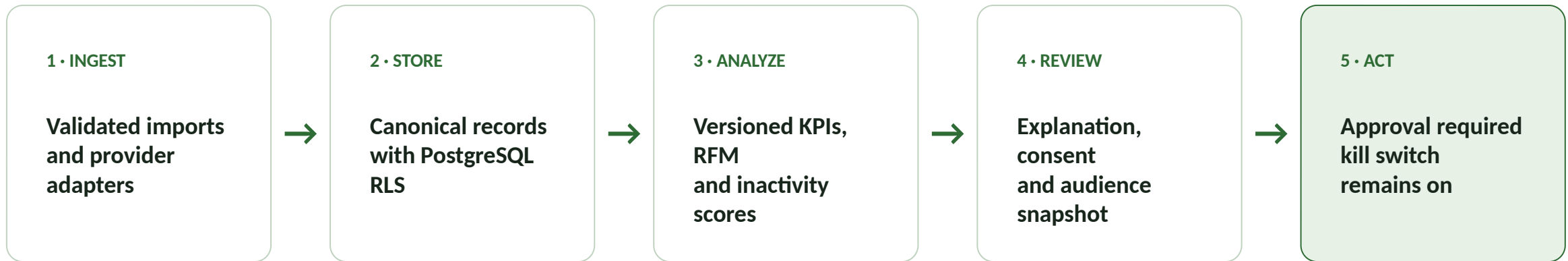
Provider metrics arrive without consistent provenance or identity.

The bottleneck is not a lack of dashboards.

It is the absence of a traceable path from data to a reviewed decision.

GrowthPilot makes provenance, tenant scope, consent and approval part of that path.

One governed workflow connects source data to human action



Controls that never drop away

Tenant scope · server-side RBAC · checksums · consent · approval · audit

The experiment uses time as a guardrail, not a random split

UCI Online Retail II

CC BY 4.0 · DOI 10.24432/C5CG6D

1,067,371

transaction lines

5,942

identified customers

90 days

future inactivity window

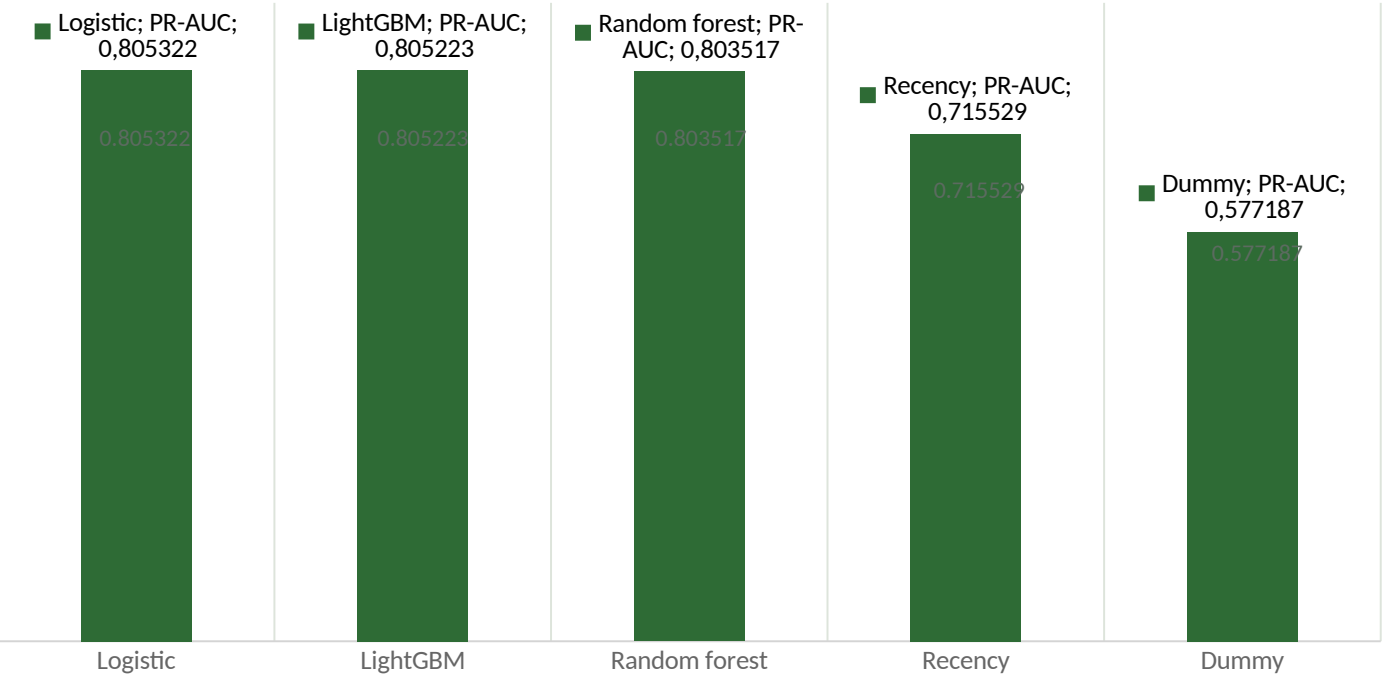
TRAIN	7 cutoffs	20,677 rows
VALIDATE	2011-03	3,349 rows
CALIBRATE	2011-06	2,659 rows
TEST	2011-09	2,772 rows

Features use events strictly before each cutoff; every label window is fully observed.

Future inactivity is an operational proxy — not contractual churn and not campaign causality.

Evidence selected a simpler model than the concept predicted

Validation PR-AUC



Selected for refinement

Regularized logistic regression

C = 0.01 improved validation PR-AUC to 0.812223. Sigmoid calibration and the ~10% threshold were then frozen before test access.

LightGBM was not chosen merely because the original concept expected it.

The frozen test shows useful ranking and visible temporal shift

0.647525

PR-AUC

0.765878

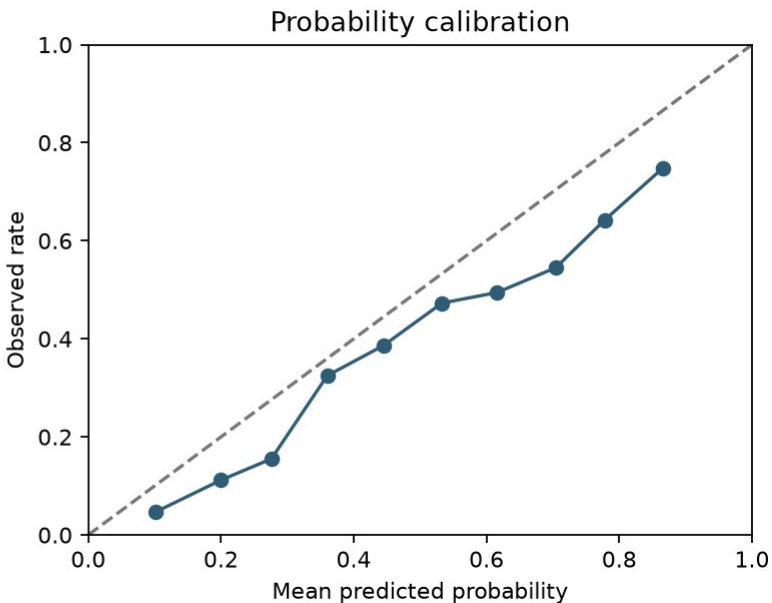
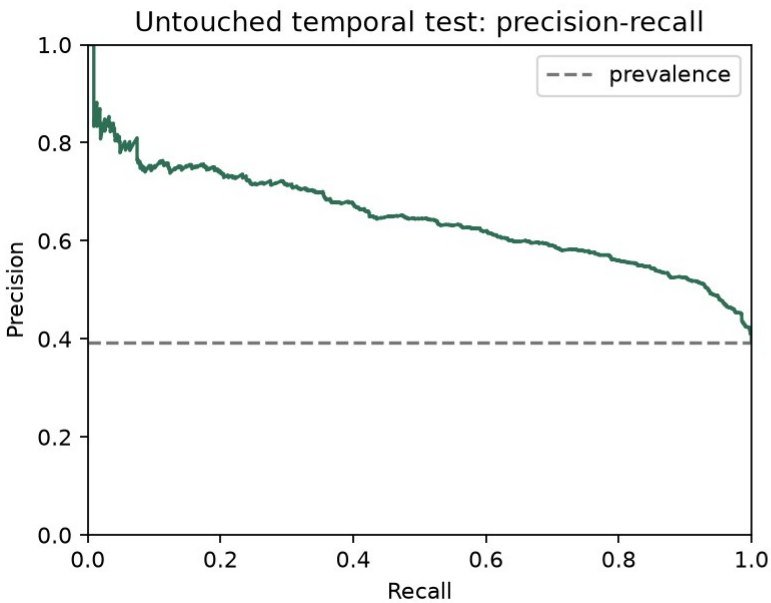
ROC-AUC

0.199854

Brier

1.904513x

Top-10% lift



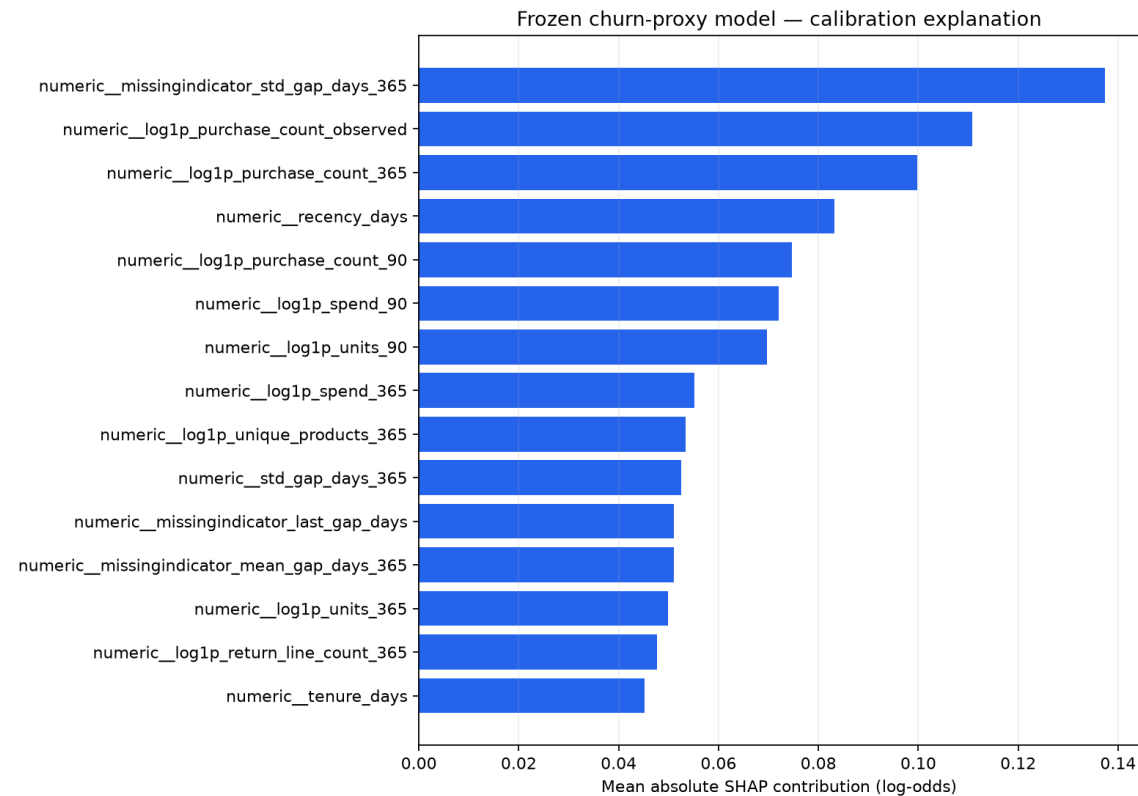
Frozen threshold

302

customers selected
10.8947% of test

Precision 0.735099
Recall 0.203857
F1 0.319195

A score can inform review, but it never authorizes an action



Validated explanation

Maximum SHAP additivity error
 8.88×10^{-16} in calibrated log-odds

- 1 Current consent
- 2 Eligible audience snapshot
- 3 Authorized reviewer
- 4 Explicit approval
- 5 Kill switch + budget cap

SHAP explains fitted score formation — not why a customer behaves or what will change them.

Local evidence is ready; production proof remains external

Completed locally

- ✓ Tenant-safe CRM / commerce / imports
- ✓ Versioned analytics and model registry
- ✓ Audiences, approvals and audit trail
- ✓ Written assignments + final presentation

External validation still required

- Live Meta / Google / LLM / OIDC sandboxes
- Browser accessibility and load validation
- Managed backup / restore rehearsal
- Randomized pilot for incremental impact

Decision requested: audit the local package. Publication, deployment and live spend remain separately gated.